Mikhail Razakov

Algorithm Engineer · Program Analysis

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Moscow / remote
from September 2026
up to 20 h/week

I build algorithmic components for libraries and compilers: 12 C++23 modules in AdvancedAlgorithms, SSA optimizations in Wist2 and conservative memory-dependence analysis in the PS-form Analyzer. I validate correctness with differential tests, exact oracles, structural invariants and sanitizers.

12 C++23 modules
AdvancedAlgorithms across graphs, trees, strings and data structures

1st place · 104/104
PS-form Analyzer: the only full-score result out of 49

AIR → SSA → AIR
SCCP-lite, folding, unreachable cleanup and DCE

EXPERIENCE

2026 — present

Creator of AdvancedAlgorithms

Header-only algorithm library · C++23

- Implemented centroid decomposition, heavy-light decomposition, LCA, Dinic, iterative Tarjan SCC, bridges and articulation points, Dijkstra, DSU, lazy segment tree, sparse table…
- Defined module contracts and complexity bounds; used differential tests against naive or reference algorithms, structural invariants and invalid-input checks.
- Added ASan/UBSan and large tests up to 500k vertices to catch stack overflow, accidental O(n^2) behaviour and ignored time limits.

July–August 2026

Engineering internship: algorithms and compiler infrastructure

MCST · LLVM compiler track · remote

- Implemented reverse postorder with cycle detection, Dijkstra shortest paths, Dinic maximum flow and Tarjan strongly connected components over a shared directed-graph model.
- Separated common graph infrastructure from individual algorithms and handled invalid input, unreachable vertices, cycles, parallel edges and disconnected components.

2024 — present

Creator and primary developer of Wist2

Open compiler/runtime platform · .NET

- Designed a text → lexer/parser/AST → bytecode/AIR → optimizations → interpreter or typed CIL backend pipeline.
- Develop modules and backend-capability contracts, semantic-equivalence checks and an experimental callable-first SSA representation.

SELECTED PROJECTS

Reusable algorithms · C++23

AdvancedAlgorithms

A header-only library of graph, tree and string algorithms plus data structures. It is positioned as a set of reusable components with explicit contracts and reproducible validation, not as copy-paste contest…

- centroid/HLD/LCA, Dinic, iterative Tarjan, bridges/articulation points, Dijkstra, DSU, lazy segment tree, sparse table, Aho–Corasick and simulated annealing.
- differential tests against Edmonds–Karp, Kosaraju, Bellman–Ford and naive oracles; structural invariants and invalid-input cases.

Program analysis · solver engineering

PS-form memory dependence analyzer

A conservative analyzer for possible overlap between parametric memory accesses in different loop iterations.

- 1st among 49 submissions and 104/104 tests.
- Normalization, ranges, modular reasoning, GCD checks, affine constraints and bounded exact enumeration.

Compiler backend · Rust

x86-64 Codegen & Register Allocation Lab

An educational backend laboratory with an SSA-like IR, liveness analysis, live intervals, a linear-scan allocator and x86-64 generation through iced-x86.

- Farthest-end spill heuristic for active live intervals.
- IR interpreter versus generated-code semantic checks.

SKILLS

RECOGNITION

First place in the technical selection
1st among 49 submissions, the only 5.0/5.0 score and the only 104/104 result in a memory-dependence analysis task.

Competitive programming
Prize winner in HSE's Vysshaya Proba olympiads in competitive and industrial programming; regional Russian informatics olympiad prize winner; winner of the Computer Science track at the…

EDUCATION

HSE University · Software Engineering · Faculty of Computer Science · bachelor's degree · Moscow · Officially admitted; full-time studies begin in September 2026. Expected graduation: 2030.